

Performance evaluation of optimized portfolio insurance strategies

Daniel Zieling*

Antje Mahayni[†]

Sven Balder[‡]

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Abstract

We use S&P 500 index return data for the time period 1985-2013 to evaluate the performance of portfolio insurance strategies. We shed light on the question if the performance of a constant proportion portfolio insurance (CPPI) strategy can be improved by means of a time-varying multiplier which depends on the estimated future volatility. Neglecting any intertemporal hedging demand, the theoretical foundation of the strategies is given by maximizing the expected utility of a HARA investor in a diffusion model setup. If the risk premium is assumed to be proportional to the variance, the optimal strategy is a CPPI strategy. Otherwise, the multiplier is time-varying (PPI). It turns out that even time-varying multiple strategies based on a rolling window of historical volatility estimates give a significant improvement of CPPI strategies. The out-performance is robust w.r.t. alternative performance measures and is also true for proportional transaction costs and adequate trigger trading.

JEL classification: G11; C51; C52; C53

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*Mercator School of Management, University of Duisburg–Essen, Lotharstr. 65, 47057 Duisburg, Germany.
E-mail: daniel.zieling@uni-due.de

[†]Mercator School of Management, University of Duisburg–Essen, Lotharstr. 65, 47057 Duisburg, Germany.
E-mail: antje.mahayni@uni-due.de

[‡]Mercator School of Management, University of Duisburg–Essen, Lotharstr. 65, 47057 Duisburg, Germany.
E-mail: sven.balder@uni-due.de

1. INTRODUCTION

The origin of portfolio insurance strategies dates back to Leland and Rubinstein (1976) and Brennan and Schwartz (1976). Leland and Rubinstein (1976) motivate the evolution of portfolio insurance with the observation that “*after the decline of 1973-74, many pension funds had withdrawn from the market only to miss the rally in 1975*” and their idea “*if only insurance were available, those funds could be attracted back to the market.*” Unsurprisingly, there are repeated revivals of portfolio insurance (PI) strategies observed on the institutional as well as the retail side of the asset management industry.¹ Along the lines of Grossman and Villa (1989) and Basak (2002), PI strategies guarantee a minimum level of wealth (at a specified time horizon), but also participate in the potential gains of a reference portfolio. To obtain a protection mechanism, a risky portfolio (or benchmark index) can be combined with a risk-free asset and/or with a suitable financial derivative. This includes dynamic versions of option-based portfolio insurance (OBPI) strategies, stop-loss strategies, buy and hold strategies and constant proportion portfolio insurance (CPPI) strategies. In the following paper, we focus on the latter approach, the CPPI approach, but also account for proportional portfolio insurance (PPI) strategies with variable multiples. In both cases, the exposure in the risky asset is proportional to the cushion. While a CPPI strategy relies on a proportionality factor which is constant throughout the investment horizon, a variable multiple strategy also allows for changes in the proportionality factor.²

Recently, there is a growing trend to implement a PPI trading rule where the multiplier is reduced in times of volatile markets and increased in less volatile times.³ Our main aim is to shed light on the question if a time-varying multiple which is based on the estimated future volatility improves the risk and return profile if compared to a constant multiple. The answer depends on the assumptions which are posed on the dynamics of the underlying risky asset (or index), i.e. the true data generating process. In consequence, we rely on S&P 500 index return data for the time period 1985-2013 and conduct a performance evaluation of PI strategies with an exogenously given guarantee constraint. For the ranking of the strategies, we consider the certainty equivalent growth rate of a HARA investor as the main criterion. We also account for the expected value growth rate and common performance measures such as the Sharpe ratio, the adjusted for skewness Sharpe ratio, the Omega measure, the Sortino ratio and the upside potential ratio.

The strategies under consideration are justified by maximizing the expected utility of an investor with hyperbolic absolute risk aversion (HARA) in a diffusion model setup. Neglecting any inter-temporal hedging demand caused by deviating from log-utility or correlated asset prices and volatilities, the strategies are optimal within a class of models

¹The practical implementation of the strategies benefits from progresses in the feasibility and effectiveness of dynamic portfolio insurance. Commercial feasibility is increased because of decreasing costs of trading and product innovations.

²The constant proportion portfolio insurance (CPPI) is introduced in Black and Jones (1987). For the basic procedure of the CPPI see also Merton (1971).

³In practice, the decision if the market scenario is considered as excessive volatile is often based on a rolling window of historical volatility. In addition (or alternatively), the decision is also linked to macro economic variables.

satisfying a specific assumption on the structure of the risk premium. From a practical point of view, a hedging demand is difficult to explain to the customer.⁴ From a theoretical point of view, we omit the hedging demand because it is strongly model depending. In particular, we do not assume that the investor maximizes her expected utility in view of one specific model. Rather, she decides on the basis of a class of models satisfying a specific assumption on the risk premium structure. The certainty equivalent growth rate optimal strategies, i.e. their myopic part, are given in terms of the Merton solution, i.e. the multiple is equal to the fraction of the current excess return and the squared volatility. If the risk premium is assumed to be proportional to the variance, the optimal strategy is a constant proportion insurance (CPPI) strategy. Otherwise, the multiple is time-varying (PPI). To be more precise, the time-varying multiple strategies are motivated by two alternative assumptions: (A1) If the risk premium is proportional to the volatility, the optimal multiple is proportional to the inverse of the volatility. (A2) If the risk premium is constant, the optimal multiple is proportional to the inverse of the squared volatility. Expecting that for some cases (regimes, respectively) it is better to take a constant multiple and for other cases (depending on market conditions) to introduce a multiple which is a non increasing function of the volatility, we also account for strategies proportional to a power of the estimated volatility where the power itself depends on the market conditions. The optimality of the *optimized* strategies is impeded by (discrete time) jumps and other market frictions such as discrete time trading and transaction costs. We account for an additional gap control in form of upper bounds on the local multipliers. Trigger trading is used to mitigate the impacts of transaction costs.

First results are presented for the raw data, i.e. the strategies are evaluated on the daily returns of non-overlapping yearly and monthly investment horizons. The observations are in favor of the time-varying multiples where the multiplier is proportional to the inverse of the historically estimated volatility (or variance). However, in view of the high dispersion caused by leveraged strategies, it is not surprising that the results are not significant. We overcome the problem of too few path observations by introducing a simulation model. According to the intuition that the answer to the problem formulation may depend on different regimes, we select and infer a regime switching EGARCH model referred to Henry (2009).

Our contributions to the existing literature are as follows. We conduct an empirical comparison of *optimized* versions of proportional portfolio insurance (PPI) strategies. Along the lines of existing empirical comparison studies, we ensure the comparability of the strategies by implementing the same guarantee for all strategies. We add to the empirical literature by focusing on time-varying multipliers. To our knowledge, we are the first ones to compare the performance of constant and variable multiplier strategies which are maximizing the expected utility of a HARA investor under different model assumptions, in particular w.r.t. the risk premium. We shed light on the question if a variable multiple, motivated by deviations from the assumption that the risk premium is proportional to the squared volatility, is able to improve the risk and return profile of a constant multiple (CPPI) strategy. It turns out that our empirical findings are in favor of volatility based multiples, and are robust w.r.t. the investment horizon, the level of risk aversion of the

⁴We are not aware of PPI products which account for a hedging demand.

HARA investor, and the choice of alternative performance measures. Theoretically, the strategies can be further improved by accounting for regime switches, i.e. by strategies which are based on a power of the estimated volatility which itself depends on the market conditions. A further contribution stems from overcoming the shortcomings due to a limited number of return observations by means of a regime switching EGARCH model. We illustrate the maximal extent to which the myopic investor can benefit from the possibility to identify the regimes. In addition, the simulation model also allows to account for gap risk, and the impacts of transaction costs and trigger trading on the performance evaluation. We illustrate the potential of trigger trading rules that control for the high turnovers. In summary, we are able to justify the industry's approach to rely on a volatility based time-varying multiplier instead of the formerly predominant CPPI approach.

Related literature concerns the topics of portfolio insurance, portfolio optimization and the impacts of transaction costs. Without postulating completeness, we refer to the following literature. The properties of continuous-time CPPI strategies are e.g. studied by Bookstaber and Langsam (2000) and Black and Perold (1992). A comparison of OBPI and CPPI is given in Bertrand and Prigent (2002a). The literature also deals with the effects of jump processes, stochastic volatility models and extreme value approaches on the CPPI method, cf. Bertrand and Prigent (2002b, 2003). Simulation and empirical studies which compare option based, zero plus underlying and constant proportion portfolio insurance (CPPI) strategies include the works of Cesari and Cremonini (2003) and Do and Faff (2004). Recent comparison studies by means of stochastic dominance criteria are Annaert et al. (2009) and Zagst and Kraus (2011). Prospect theory is used in Dierkes et al. (2010) and Dichtl and Drobetz (2010). Bertrand and Prigent (2011) rely on downside risk measures, in particular the Omega performance measure, to compare portfolio insurance strategies. Balder et al. (2012) point out some pitfalls of time invariant (TIPP) strategies immanent in index products and individual (iCPPI) strategies which are caused by horizon effects. An analysis of gap risk, i.e. the risk that the guarantee is violated, is provided in Cont and Tankov (2009) and Balder et al. (2009). Cont and Tankov (2009) introduce the gap risk by considering jump-diffusion models. Balder et al. (2009) introduce the gap risk by adding trading restrictions such that the analysis also captures the effects of transaction costs. Accounting for gap risk can also motivate variable multiple strategies. Thus, our paper is also linked to the literature focusing on additional constraints such as VaR and ES constraints, cf. Herold et al. (2005), Hamidi et al. (2009a,b), Jiang et al. (2009), Ben Ameur and Prigent (2011) and Ho et al. (2011). In the context of portfolio insurance strategies, trigger trading is already suggested in Black and Perold (1992) and Boulier and Kanniganti (1995). Here, the portfolio is rebalanced when the price of the risky asset exceeds an upper or lower return, i.e. the strategy is a buy and hold strategy as long as the asset price increment is not too extreme in terms of the trading filter. More recent literature poses the trigger on the implicit multiplier (or implicit exposure, respectively) of the CPPI, cf. Paulot and Lacroze (2009), Hamidi et al. (2009a), MKaouar and Prigent (2007), Jessen (2009) and Khuman and Constantinou (2009). Concerning the literature on portfolio planning, references include Merton (1971) who solves the portfolio planning problem for a HARA investor with inter-temporal consumption. Amongst others, Kim and Omberg (1996) and Barberis (2000) consider optimal portfolios when the stock returns are predictable. Stochastic volatility and jumps are, for example, considered in Liu et al. (2003), Branger et al. (2008) and Muck (2010). Basak (2002) shows that adding

a subsistence level to the problem leads to portfolio insurance strategies. Grossman and Zhou (1993) show that the time invariant portfolio insurance (TIPP) strategy is the optimal investment policy under drawdown constraints.

The outline of the paper is as follows. Section 2 reviews some theoretical optimization results and defines benchmark assumptions on the risk premium. Along the ways, we introduce a regime switching model/strategy where the regimes correspond to the benchmark assumptions on the risk premium. In Section 3, we describe the data and discuss some rather descriptive results concerning the evaluation of constant proportion portfolio insurance strategies and the time-varying multiple strategies proportional to the inverse of the historically estimated volatility and variance, respectively. To overcome the shortcomings of the limited number of return observations, we select and infer a simulation model in Section 4. Using simulated paths, we evaluate and rank various CPPI and PPI strategies in Section 5. In particular, we also allow for switching strategies. In addition, we account for gap risk and transaction costs. Besides daily trading, we implement a trading filter which mitigates the negative effects caused by the transaction costs. Section 6 concludes the paper.

2. BASIC OPTIMIZATION RESULTS AND STRATEGIES UNDER CONSIDERATION

The following section gives the foundations of our research question whether, when, and to what extent it is possible to improve CPPI strategies by means of dynamic (but simple) PPI strategies. In the first instance we recall the associated optimization problems which justify the strategies as optimal. Then, we account for deviations from these model assumptions and market frictions such as discrete time trading and transaction costs which impede the optimality of the strategies. Along the ways, we discuss why it is important to concentrate on tractable and simple to implement strategies in favor of more sophisticated but strongly model dependent strategies.

2.1. The investment problem. Throughout the following, optimality is defined in terms of an investor with hyperbolic absolute risk aversion (HARA). The investor maximizes her (expected) utility u^{HARA} of terminal wealth where

$$u^{\text{HARA}}(x) = \begin{cases} \frac{(x-G)^{1-\gamma}}{1-\gamma} & \text{for } \gamma > 0 \text{ and } \gamma \neq 1 \\ \ln(x-G) & \text{for } \gamma = 1. \end{cases} \quad (1)$$

The guarantee G ($G \geq 0$) defines her subsistence level. A terminal wealth below the guarantee G is prohibitive. In addition to G , the risk aversion depends on γ . For $G = 0$, the level of relative risk aversion is equal to γ , i.e. utility functions with constant relative risk aversion (CRRA) also belong to the class of HARA utility functions. However, we consider portfolio insurers, i.e. we assume $G > 0$. The relative risk aversion is $\gamma x(x-G)^{-1}$. It is decreasing in wealth and converges to γ , i.e. $\lim_{x \rightarrow \infty} \gamma x(x-G)^{-1} = \gamma$. For $x > G$ it is larger than γ , i.e. holds $\gamma x(x-G)^{-1} > \gamma$. Ceteris paribus, the relative risk aversion is the higher the higher the guarantee G is.⁵ There are two investment possibilities: a risky asset S and

⁵For a CRRA investor ($G = 0$), empirical research supports a level of risk aversion larger than one, i.e. $\gamma > 1$. Recently, Chiappori and Paiella (2011) use panel data to analyze how individuals' portfolio allocation between risky and riskless assets varies in response to changes in total financial wealth. Their result supports the CRRA assumption. In particular, the authors estimate a median level of risk aversion

a (locally) risk-free money market account. For reasons of clarity and comprehensibility we restrict the discussion on excess returns such that the risk-free rate is zero. If not mentioned otherwise, all stochastic processes under consideration are continuous time semi-martingales defined on a stochastic basis $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T^*]}, P)$ which satisfies the usual hypotheses. A continuous-time investment strategy or saving plan for the interval $[0, T]$ can be represented by a predictable process $(\phi_t^{(S)}, \phi_t^{(B)})_{t \in [0, T]}$ such that $\phi^{(S)}$ is an admissible integrand for S and $\phi^{(B)}$ is an admissible integrand for all $t \in [0, T]$. At t , $\phi_t^{(S)}$ denotes the number which is invested in the risky asset S and $\phi_t^{(B)}$ denotes the amount which is put into the money market account. The corresponding value process is denoted by $(V_t)_{t \in [0, T]}$, i.e.

$$V_t = \phi_t^{(S)} S_t + \phi_t^{(B)}.$$

The strategy is self-financing if, after an initial investment $V_0 = v_0$, there are no further in- or outflows of funds, i.e.

$$dV_t = \phi_t^{(S)} dS_t + \phi_t^{(B)} dt \text{ where } V_0 = v_0.$$

The investor now seeks to optimize her expected utility over all self-financing strategies starting with her initial wealth v_0 and the terminal restriction posed by $V_T \geq G$ a.s.. Alternatively to the number of shares, the (self-financing) strategies can be stated in terms of the fraction $\tilde{m}_t$ of wealth invested in the risky asset S while the remaining portfolio fraction is left for the risk-free investment. In particular, $\tilde{m}_t = \frac{E_t}{V_t}$, where $E_t = \phi_t^{(S)} S_t$ denotes the time t asset exposure. An investment rule which states the asset exposure E_t proportionally to the portfolio value V_t is called a proportional strategy. The proportionality factor $\tilde{m}_t$ is called multiplier. A constant proportional strategy is implied by a constant multiplier $\tilde{m}_t = \tilde{m}$, for all $t \in [0, T]$. If there are no further restrictions than the usual integrability and predictability conditions, the class of proportional strategies is equal to the set of admissible strategies. If $\tilde{m}_t > 1$ (for all $t \in [0, T]$), the trading rule prescribes a reduction (increase) of the risky exposure in the case of downward (upward) moving markets. This is also called momentum strategy. The postulation of a subsistence level G ($G < V_0$) implies that the optimal strategy is a (proportional) portfolio insurance (PPI) strategy. Here, the proportionality factor m_t is applied to the difference of the portfolio value V_t and the floor (guarantee, respectively) G , i.e.⁶

$$E_t = m_t \times (V_t - G) = m_t \times C_t$$

where $C_t := V_t - G$ denotes the cushion at time t . The (initial) guarantee is given by a fraction $\alpha \in [0, 1]$ of the initial portfolio value, i.e. $G = \alpha V_0$, ensuring $C_0 > 0$. In the following, it is convenient to represent the expected utility maximization problem in terms of the proportionality factor m , i.e.

of $\gamma = 1.7$. For $G > 0$, it is difficult to give empirical support for the HARA assumption. However, assuming that the subsistence level G is exogenously given may well imply a rather low γ (e.g. $\gamma = 1$), i.e. if G is rather high compared to the initial investment, the drift and the time horizon under consideration.

⁶A practical advantage of the CPPI (a constant multiplier m) or a simple rule based multiplier is the simplicity of the strategy. In this case, the PPI method is less demanding than synthesizing an option payoff, i.e. the dynamic version of option based portfolio protection (OBPI). More importantly, the PPI method can, theoretically, be applied to an arbitrary (in particular, an infinite) investment horizon T .

$$\begin{aligned} & \max_m E [u^{\text{HARA}}(V_T)] \text{ s.t. } V_T \geq G \\ & = \max_m E [u^{\text{CRRA}}(C_T)] \end{aligned} \quad (2)$$

where $(m_t)_{0 \leq t \leq T}$ is restricted by the set of predictable and square integrable processes. u^{CRRA} denotes the CRRA utility function which is defined by u^{HARA} in combination with $G = 0$, cf. Equation (1). In particular, the maximization problem of the HARA investor is represented by the maximization problem of a CRRA investor who measures her utility in terms of the cushion C instead of the value V . Alternatively, the optimization problem can be stated in terms of the certainty equivalent (CE) growth rate comprising the same information as the expected utility. Since the CE growth rate is more convenient to interpret, the empirical evaluation results are stated in CE terms. The CE growth rate y^{CE} is defined by

$$\begin{aligned} u^{\text{HARA}}(V_0 e^{y^{\text{CE}} T}) &= E [u^{\text{HARA}}(V_T)] \Leftrightarrow u^{\text{CRRA}}(C_0 e^{y^{\text{CE}} T}) = E [u^{\text{CRRA}}(C_T)], \\ \text{i.e. } y^{\text{CE}} &= \begin{cases} \frac{1}{T} \ln \left(E \left[\left(\frac{C_T}{C_0} \right)^{1-\gamma} \right]^{\frac{1}{1-\gamma}} \right) & \text{for } \gamma > 0 \text{ and } \gamma \neq 1 \\ \frac{1}{T} E \left[\ln \frac{C_T}{C_0} \right] & \text{for } \gamma = 1. \end{cases} \end{aligned} \quad (3)$$

2.2. Model dependent optimal investment policies. The solution of the optimization problem (2) is well known in a Black and Scholes model setup where the price dynamics of the risky asset are given by a geometric Brownian motion with constant drift μ (equity risk premium λ , respectively) and constant volatility σ , and which does not account for market frictions such as trading restrictions and transaction costs. Under the assumption of independent and identically distributed asset price increments it is straightforward to show that the optimal multiplier is given by $m^* = \lambda/(\gamma\sigma^2)$.⁷ Obviously, the empirical data is not consistent with the assumption of the Black and Scholes model. Assume for now that the risky asset dynamics S are described by a diffusion setup for the risky asset and a diffusion setup for its volatility, i.e.

$$dS_t = S_t(\lambda_t dt + \sigma_t dW_t^S) \quad (4)$$

$$d\sigma_t = \alpha_t dt + \beta_t dW_t^\sigma \quad (5)$$

where $(W_t^S)_{0 \leq t \leq T}$ and $(W_t^\sigma)_{0 \leq t \leq T}$ each denote a one dimensional standard Brownian motion and λ_t denotes the (possibly stochastic) equity risk premium. It is assumed that all coefficients of the above SDEs are progressively measurable with respect to the filtration $(\mathcal{F}_t)_{0 \leq t \leq T}$. Furthermore, the two Brownian motions may be correlated. For the sake of simplicity, we assume a constant correlation denoted by ρ , i.e. the quadratic variation at t ($t \in [0, T]$) is given by $[W^S, W^\sigma]_t = \rho t$.

⁷The above result is well known. Amongst other results, Merton (1971) showed that the optimal fraction of wealth m^* which is invested in the risky asset for a CRRA investor described by a utility function of the form $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$ for $\gamma > 0$ and $\gamma \neq 1$, and $u(x) = \ln x$ for $\gamma = 1$, is given by $m^* = \frac{\lambda}{\gamma\sigma^2}$. The optimality of a CPPI with multiplier $m = \frac{\lambda}{\gamma\sigma^2}$ associated with the introduction of a subsistence level G into the utility function, i.e. a more general HARA utility function, is for example considered in Basak (2002).

For predictable and square integrable processes $(m_t)_{0 \leq t \leq T}$, the cushion dynamics is given by

$$dC_t = C_t m_t \frac{dS_t}{S_t} = C_t (m_t \lambda_t dt + m_t \sigma_t dW_t^S).$$

In particular, it holds

$$\begin{aligned} u^{\text{CRRA}}(C_T) &= u^{\text{CRRA}} \left(C_0 e^{\int_0^T (m_u \lambda_u - \frac{1}{2} (m_u \sigma_u)^2) du + \int_0^T m_u \sigma_u dW_u^S} \right) \\ &= \begin{cases} \frac{1}{1-\gamma} C_0^{1-\gamma} e^{\int_0^T (1-\gamma)(m_u \lambda_u - \frac{1}{2} (m_u \sigma_u)^2) du + \int_0^T (1-\gamma) m_u \sigma_u dW_u^S} & \text{for } \gamma > 1 \\ \ln C_0 + \int_0^T (m_u \lambda_u - \frac{1}{2} (m_u \sigma_u)^2) du + \int_0^T m_u \sigma_u dW_u^S & \text{for } \gamma = 1. \end{cases} \end{aligned}$$

It is well known that the strategy for an expected-utility-maximizing investor is myopic for $\gamma = 1$ or for all $\gamma \geq 1$ in combination with $\rho = 0$.⁸ The myopic investor maximizes her expected incremental utility $E[du(C_t)]$. Using Itô's Lemma it is straightforward to show

$$\begin{aligned} E[du(C_t)|\mathcal{F}_t] &= E \left[\frac{\partial u(C_t)}{\partial C_t} dC_t + \frac{1}{2} \frac{\partial^2 u(C_t)}{\partial C_t^2} d[C]_t \middle| \mathcal{F}_t \right] \\ &= C_t^{\gamma-1} \left(m_t \lambda_t dt - \frac{1}{2} \gamma m_t^2 \sigma_t^2 \right) dt. \end{aligned}$$

The first-order condition immediately implies that the optimal (myopic) multiplier $m_t^{*,\text{sv}}$ is given by

$$m_t^{*,\text{sv}} = \frac{\lambda_t}{\gamma \sigma_t^2} \quad \text{for all } t \in [0, T]. \quad (6)$$

In particular, the integrability conditions are satisfied if $\int_0^T |\lambda_t m_t^{*,\text{sv}}| + (\sigma_t m_t^{*,\text{sv}})^2 dt < \infty$, which corresponds to $\int_0^T \left(\frac{\lambda_t}{\sigma_t} \right)^2 dt$.⁹

In general, there is an inter-temporal hedging demand for non-logarithmic utility ($\gamma \neq 1$), i.e. if $\rho \neq 0$. In particular, a time depending hedging demand immediately implies that a time-varying multiplier is able to improve the performance of a constant multiple. However, the strategies under consideration only account for an optimized myopic part, i.e. we neglect the strongly model dependent hedging demand. We do not assume that the investor maximizes her expected utility in view of one specific model – based on an accurate specification of the volatility and correlation processes – but rather for a whole class of models.¹⁰ To be more precise, we consider the question whether a time-varying multiple (myopic demand, respectively) is able to improve the results of a constant multiple immanent in a CPPI strategy. Neglecting any hedging demand, the answer only depends on the assumption on the link between the local risk premium λ_t of the risky asset

⁸Cf. for example Merton (1973). An optimal portfolio strategy is called myopic if the optimal decisions of a long-term investor and an otherwise identical short-term investor coincide. For a detailed discussion of myopic portfolio choices we refer to Campbell and Viceira (2002).

⁹The above condition is usually satisfied for standard financial markets as defined by Karatzas and Shreve (1999). For the empirical comparison study, the PPI strategies are based on the realized variance. Along the lines of Schied (2013), such strategies are also well-defined if the above condition is not met.

¹⁰For a similar reasoning we also refer to Diris et al. (2013).

and the local volatility σ_t . For example, the theoretical literature about portfolio planning often relies on the assumption that the risk premium is proportional to the variance, i.e.

$$(A0) \quad \lambda_t = \bar{\lambda} \sigma_t^2.$$

In particular, (A0) implies that the Sharpe ratio is increasing in the volatility. Alternative assumptions are that the risk premium is proportional to the volatility (A1) or is constant (A2), i.e.

$$(A1) \quad \lambda_t = \bar{\lambda} \sigma_t, \quad (A2) \quad \lambda_t = \bar{\lambda}.$$

In consequence, we obtain the following implications for the performance evaluation of variable multiple strategies: PPI strategies can not outperform the optimal constant multiple under the assumption (A0). However, they can outperform the CPPI if either the Sharpe ratio is constant (A1) or if the risk premium is constant (A2). Since the answer may also depend on different regimes, we account for the possibility that there are (depending on the market conditions) regimes which may correspond to the above assumptions posed on the risk premium. Theoretically, this can be modeled by a finite state Markov chain $(X_t)_{0 \leq t \leq T}$ (adapted to the information structure) with state space $\{1, \dots, N\}$. For $c = (c_1, \dots, c_N)'$ and $\alpha = (\alpha_1, \dots, \alpha_N)'$, we set

$$\lambda_t = c_{X_t} \sigma_t^{\alpha_{X_t}}.$$

For $N = 3$, $c = (c_1, c_2, c_3)'$, and $\alpha = (\alpha_1, \alpha_2, \alpha_3)' = (2, 1, 0)'$, the regimes are consistent with the three assumptions on the risk premium. In particular, regime 1 corresponds to the assumption (A0), regime 2 to (A1), and regime 3 to (A2). However, different values of α or other specifications of λ_t are able to capture more general regimes as well. If the Markov chain is observable it is straightforward to show that the myopic part of the multiplier satisfies

$$m_t^{*,\text{switch}} = c_{X_t} \sigma_t^{\alpha_{X_t} - 2}. \quad (7)$$

2.3. Introduction of gap risk and gap risk control. The risk that the guarantee is not achieved by a dynamic portfolio insurance strategy is called gap risk. In contrast to the above theoretical model setup, gap risk is introduced by accounting for (discrete time) jumps in the asset price. Intuitively, it is clear that gap risk is linked to the inadequacy to omit the risky positions *fast enough* in downward moving markets. A formal proof that there are no other sources for gap risk apart from jumps in asset prices or discrete rebalancing is given in Schied (2013).

Obviously, the introduction of gap risk impedes the optimality of the above strategies.¹¹ Similar to the reasoning as with regard to the inter-temporal hedging demand caused by deviations from $\gamma = 1$ in combination with a non zero asset and volatility correlation, we view it beyond the scope of the investor to accurately specify a jump-diffusion process for the asset dynamics. In view of model risk and practical applications under market frictions, it is thus interesting to analyze if one of the above simple and tractable to implement strategies is preferred by the investor. Nevertheless, it is important to account

¹¹Notice that jumps thus impede the concept of dynamic portfolio insurance. Accounting for a positive jump probability (without restrictions on the downside jump sizes and multipliers) introduces gap risk such that a dynamic PPI strategy is not optimal.

for a meaningful gap control and an adequate trigger design. We start with the gap risk and discuss the implications of transaction costs afterwards.

A first insight on the impact of gap risk can be obtained by considering the sensitivities of the strategies to the introduction of a gap risk control. Since the strategies are implemented in discrete time, we represent the gap event in a discrete time setup. At the same time this allows to account for (real) jumps, i.e. jumps which are not exclusively introduced by trading restrictions in form of discrete time trading. Therefore, let τ^n denote a sequence of equidistant refinements of the interval $[0, T]$, i.e. $\tau^n = \{t_0^n = 0 < t_1^n < \dots < t_{n-1}^n < t_n^n = T\}$, where $t_{k+1}^n - t_k^n = \frac{T}{n}$ for $k = 0, \dots, n-1$. To simplify the notation, we drop the superscript n and denote the set of trading dates with τ instead of τ^n . The restriction that trading is only possible immediately after $t_k \in \tau$ implies that the number of shares held in the risky asset is constant on the intervals $]t_k, t_{k+1}]$ for $k = 0, \dots, n-1$. However, the fractions of wealth which are invested in the assets change as assets prices fluctuate. Thus, it is necessary to consider the number of shares held in the risky asset $\phi^{(S)}$ and the number of bonds (position in the money account, respectively) $\phi^{(B)}$. Along the lines of Balder et al. (2009), we call a strategy $\phi^\tau = (\phi^{(S),\tau}, \phi^{(B),\tau})$ discrete-time PPI if for $t \in]t_k, t_{k+1}]$ and $k = 0, \dots, n-1$

$$\phi_t^{(S),\tau} := \max \left\{ \frac{m_{t_k} C_{t_k}^\tau}{S_{t_k}}, 0 \right\}, \quad \phi_t^{(B),\tau} := V_{t_k}^\tau - \eta_t^\tau S_{t_k}.$$

It is straightforward to show that the cushion dynamics are given by

$$C_{t_{k+1}}^\tau = C_{t_0}^\tau \prod_{i=1}^{\min\{\nu, k+1\}} \left(m_{t_{i-1}} \frac{S_{t_i}}{S_{t_{i-1}}} - (m_{t_{i-1}} - 1) \right),$$

where $\nu := \min \{t_k \in \tau | V_{t_k}^\tau(\alpha) - G \leq 0\}$, and $\nu = \infty$ if the minimum is not attained. Consistent with the perception of a myopic decision rule, we consider a local quantile guarantee condition posed by¹²

$$P_{t_k} (C_{t_{k+1}} > 0 | C_{t_k} > 0) \geq 1 - \epsilon,$$

where ϵ denotes some exogenously specified upper bound on the local shortfall probability. Together with the observation that

$$P_{t_k} (C_{t_{k+1}} > 0 | C_{t_k} > 0) = P_{t_k} \left(m_{t_k} \frac{S_{t_{k+1}}}{S_{t_k}} - (m_{t_k} - 1), > 0 \right)$$

this implies an upper bound $\bar{m}_{t_k}$ on the admissible multiplier, i.e. a gap control affords to limit the multiplier at t_k by $\bar{m}_{t_k}$. Let $\tilde{F}_{t_k}$ denote the marginal distribution function of the standardized simple return $\frac{S_{t_{k+1}}}{S_{t_k}} - 1$. The upper bound is given by

$$\bar{m}_{t_k} = \left| E_{t_k} \left[\frac{S_{t_{k+1}}}{S_{t_k}} \right] - 1 + \sqrt{\text{Var}_{t_k} \left[\frac{S_{t_{k+1}}}{S_{t_k}} \right]} \tilde{F}_{t_k}^{-1}(\alpha) \right|^{-1}.$$

For example, the above condition is used in Hamidi et al. (2009c) who estimate the conditional (upper) multiplier by means of Dynamic AutoRegressive Expectile (DARE)

¹²Obviously, there are other meaningful criterions to control the gap risk, e.g. to control the expected shortfall as well as the global gap probability. However, besides contradicting the idea of a myopic investor this is beyond the scope of this paper.

Summary and test statistics of daily (yearly) excess returns

Mean excess return	0.000229	(0.061206)	
Standard deviation	0.011508	(0.185557)	
Skewness	-0.841247		
Kurtosis	24.962414		
Minimum	-0.204590		
Maximum	0.115778		
	t-statistic	critical value	p-value
Skewness	-29.5835	-3.2905	0.0000
Kurtosis	386.2201	3.2905	0.0000
Normality (Jarque-Bera)	149,940	14.6257	0.0000

TABLE 1. Summary statistics of the whole data set containing 7,418 observations. If convenient, the corresponding yearly values are included in brackets. In particular, $\bar{\lambda}_{\text{annual}} = (1 + \bar{\lambda}_{\text{daily}})^{260} - 1$ and $\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{260}$. Critical values are reported for $\alpha = 0.1\%$.

models. In view of the *optimized* multiples, it is worth to emphasize that the upper limit implied by $\bar{m}_{t_k}$ is less binding for strategies which already reduce the multiple in times of higher volatility.

Finally, we comment on a further challenge on the implementation of CPPI and dynamic PPI strategies, i.e. transactions costs. Notice that because of the guarantee constraint, it is not possible to change the risk-free position of the PPI method (which is needed without transaction costs). Rather, the transaction costs are to be financed by a reduction of the asset exposure arising in the case without transaction costs.¹³ Along the lines of Balder et al. (2009), the discrete time cushion process under proportional transaction costs with proportionality factor θ is given by

$$C_{t_k+} = C_{t_k} - \theta \left| m_{t_k} C_{t_k+} - m_{t_{k-1}} C_{(t_{k-1})+} \frac{S_{t_k}}{S_{t_{k-1}}} \right|, \quad (8)$$

i.e. $m_{t_k} C_{t_k+}$ denotes the asset exposure immediately after a transaction cost adjustment. Obviously, accounting for transaction costs implies a further reduction of the upper bound on an admissible multiplier with regard to gap risk, cf. Balder et al. (2009). From a practical point of view, the above reasoning emphasizes the importance to consider simple and rule based strategies which can be tractably adjusted to account for gap risk and transaction costs, i.e. to determine an adequate trading filter.

3. PERFORMANCE EVALUATION

We consider a data set containing index return data of the S&P 500 (composite index) for the past 28 years and about 5 month. The data is provided by Thomson Reuters for the time period 2 January 1985 - 7 June 2013. The composite index is net of dividends. The loss of dividends is interpreted as the management fee for the PPI products.¹⁴ The

¹³Black and Perold (1992) assume that rebalancing occurs *net of transaction costs*, too.

¹⁴This assumption is e.g. also posed in Bertrand and Prigent (2011).

data contains daily simple returns. The number of observations is 7,418. This resembles approximately 260 trading days per year. The daily simple return is denoted by r_t^S , i.e. $r_t^S = \frac{S_t}{S_{t-1}} - 1$. As a proxy for the (locally) risk-free rates, we use discount yields of T-Bills with 91 days to maturity. In particular, interest rate data congruent to the S&P 500 returns is used for the discounting. The whole analysis is then conducted w.r.t. the simple excess returns. To be more precise, let r_t^f denote the risk-free daily interest rate. The excess return is then given by $R_t = r_t^S - r_t^f$. A summary and test statistics accounting of the whole time period is provided in Table 1. The mean daily excess return is $\hat{\lambda} = 0.000229$ or about 0.0612 per year. The skewness is negative and significant. There is also significant excess kurtosis, i.e. extremes tend to be more pronounced than for a normally distributed random variable. A Jarque-Bera test confirms the departure from a normal distribution.

To get some first intuition about the performance of *optimized* CPPI and time-varying strategies, we consider a first set of benchmark strategies. The initial investment is $V_0 = 100$ and the guarantee is $G = 90$. The investment horizon is $T = 1$ ($T = \frac{1}{2}$, respectively). Recall that the guarantee also defines the subsistence level of the utility function u^{HARA} , cf. Equation (1). The investor can be assessed as highly risk averse because of the rather high guarantee such that a realistic choice of γ implies a rather low γ . If not mentioned otherwise, we use $\gamma = 1$. Since this resembles a log-utility measured on the cushion there is no inter-temporal hedging demand. Comparing the outcomes of PPI strategies which are solely based on a myopic component has the advantage that there are no distortion effects (utility losses, respectively) due to an inter-temporal hedging demand. However, since log-utility implies a very specific attitude towards risk we consider the impact of deviations from $\gamma = 1$ in the subsequent simulation study. For now, we evaluate the growth rates of some representative strategies for non-overlapping investment horizons. As a proxy for the optimal constant multiplier consistent with assumption (A0), we rely on the long term estimates for the risk premium $\hat{\lambda}$ and the volatility $\hat{\sigma}$, i.e.

$$m^{*,\text{const}} = \frac{\hat{\lambda}}{\hat{\sigma}^2} = \frac{0.000229}{0.011508^2} = 1.7256.$$

As additional benchmark strategies we also include a CPPI strategy with $m = 1$ (static portfolio insurance), $m = 4$ and $m = 6$. Motivated by the (alternative) assumptions (A1) and (A2), we also include strategies which are proportional to the inverse of a historically estimated volatility (variance).¹⁵ According to the assumptions posed on the risk premium,

¹⁵For the variable multiplier strategy m_t , we implement different methods for estimating the local volatility σ_t . Our main focus is on a non-parametric approach. We rely on historical volatility estimates $\hat{\sigma}_t = \sigma_{t,xM}$, where $x \in \{1, 2\}$ denotes the number of months of the estimation window. For example, $\sigma_{t,1M}$ is calculated using a window of the latest 21 daily excess returns observed at days $(t, t-1, \dots, t-20)$. The results are only reported for the better performing strategies which are based on a one month window. It is worth mentioning that a further meaningful set of strategies can be based on implicit volatilities derived from observed option prices. Implicit volatilities are often considered as more informative. We omit such strategies because of the following reason. Unsurprisingly, the performance results obtained for the limited number of observations do not allow for a significant comparison. In a next step, we thus infer a simulation model, a regime switching EGARCH model, for the excess returns under the *real world* probability measure. A consistent modelling of the risk neutral dynamics is beyond the scope of the paper.

Mean yearly growth rates of selected PPI strategies

Panel A: Investment horizon $T = 1$ year

	$\frac{1}{T}E[\ln \frac{C_T}{C_0}]$	$\frac{1}{T}E[\ln \frac{V_T}{V_0}]$	$E[V_T]$	$\min V_T$	SR	$ASSR$	$\Omega - 1$	SoR	UPR
$m_{t,(A1)}$	0.075 <i>0.362</i>	0.014 <i>0.043</i>	101.496 <i>4.502</i>	95.454	0.332	0.391	1.710	0.876	1.389
$m_{t,(A2)}$	0.077 <i>0.700</i>	0.038 <i>0.166</i>	105.695 <i>25.723</i>	93.787	0.221	0.289	3.628	2.146	2.737
$m = 1$	0.032 <i>0.165</i>	0.004 <i>0.016</i>	100.457 <i>1.574</i>	96.313	0.290	0.268	1.062	0.461	0.896
$m^{*,\text{const}}$	0.034 <i>0.299</i>	0.007 <i>0.027</i>	100.745 <i>2.715</i>	94.066	0.274	0.259	0.966	0.446	0.908
$m = 4$	-0.100 <i>0.839</i>	0.012 <i>0.061</i>	101.413 <i>6.145</i>	90.553	0.230	0.232	0.771	0.410	0.942
$m = 6$	(-0.278) <i>1.401</i>	0.013 <i>0.088</i>	101.683 <i>9.176</i>	87.622 1	0.183	0.193	0.641	0.364	0.931

Panel B: Short investment horizon $T = 1/12$ year

$m_{t,(A1)}$	0.076 <i>0.344</i>	0.013 <i>0.034</i>	100.112 <i>0.984</i>	96.118	0.114	0.114	0.332	0.175	0.701
$m_{t,(A2)}$	0.085 <i>0.616</i>	0.026 <i>0.063</i>	100.230 <i>1.834</i>	94.698	0.126	0.130	0.425	0.221	0.742
$m = 1$	0.030 <i>0.176</i>	0.004 <i>0.017</i>	100.038 <i>0.495</i>	97.213	0.076	0.074	0.234	0.104	0.551
$m^{*,\text{const}}$	0.031 <i>0.311</i>	0.007 <i>0.030</i>	100.064 <i>0.852</i>	95.377	0.075	0.074	0.230	0.105	0.560
$m = 4$	-0.098 <i>0.818</i>	0.015 <i>0.068</i>	100.148 <i>1.955</i>	91.045	0.075	0.076	0.229	0.111	0.595
$m = 6$	(-0.220) <i>1.108</i>	0.022 <i>0.100</i>	100.226 <i>2.907</i>	89.074 1	0.075	0.079	0.235	0.120	0.630

TABLE 2. Mean performance (*and standard deviations*) for selected strategies over the past 28 years. For Panel A and B the investment horizon is $T = 1$ and $T = \frac{1}{12}$ year, respectively. The initial portfolio value is given by $V_0 = 100$. The initial guarantee is $G = 90$. The investment quote is bounded to a maximum of 200% of the current portfolio value V_t . The optimal constant multiplier is $m^{*,\text{const}} = 1.7256$. The expected growth rates presented in the first column for the strategy $m = 6$ are given in brackets. This strategy failed to ensure the guarantee of 90 (also compare the “ $\min V_T$ ” column). In this case the cushion is negative and the expected cushion growth rate is equal to $-\infty$. Since this result is less informative, the quantity in parentheses reports the expected cushion growth rate if the negative cushion years (month) are simply excluded. However, they cannot be compared to the other growth rates because of the introduced bias. The boldface number in the second line of the $\min V_T$ column counts the number of floor violations.

we denote these strategies with $m_{t,A1}$ and $m_{t,A2}$ where

$$m_{t,A1} = \frac{\hat{\lambda}}{\hat{\sigma}} \frac{1}{\sigma_{t,1M}}, \quad (9)$$

$$m_{t,A2} = \hat{\lambda} \frac{1}{\sigma_{t,1M}^2}. \quad (10)$$

Performance measures	
Sharpe ratio (SR)	$\frac{E[V_T - V_0 e^{rT}]}{\sqrt{\text{Var}[V_T]}}$
Adjusted Sharpe ratio ($ASSR$)	$SR \sqrt{1 + b_3 \frac{\text{Skew}}{3} SR}$ where $b_3 = 2$
Omega measure (Ω)	$\frac{E[\max\{V_T - K, 0\}]}{E[\max\{K - V_T, 0\}]}$
Sortino ratio (SoR)	$\frac{E[V_T - K]}{\sqrt{E[(\max\{K - V_T, 0\})^2]}}$
Upside potential ratio (UPR)	$\frac{E[\max\{V_T - K, 0\}]}{\sqrt{E[(\max\{K - V_T, 0\})^2]}}$

TABLE 3. Summary of performance measures.

$\sigma_{t,1M}$ is the volatility estimate calculated using a window of the latest 21 daily excess returns observed at days $(t, t-1, \dots, t-20)$. Theoretically, a PPI strategy may imply that the asset exposure is financed by an unlimited position in the risk-free asset. In contrast, traded PPI products include a cap on the investment fractions. Therefore, we focus on strategies with a capped investment fraction.¹⁶

The descriptive statistics w.r.t. yearly and monthly investment horizons are summarized in Table 2. All observations are based on non-overlapping investment periods such that the number of observations is 28 (28×12 plus 5 month in 2013) in the case of a yearly (monthly) investment horizon. In addition to the average cushion growth rates (certainty equivalent growth rates, respectively), Table 2 summarizes the average rate of return, the average payoff, the Sharpe ratio SR (cf. Sharpe (1966)), the adjusted for skewness Sharpe ratio $ASSR$ (cf. Zakamouline and Koekebakker (2009)), the Omega measure Ω (cf. Keating and Shadwick (2002)), the Sortino ratio SR (cf. Sortino and Price (1994)), and the upside potential ratio UPR . The definitions of the performance measures are given in Table 3. Throughout the following, we use the standard versions of the performance measures which rely on the choice $K = V_0$.

First, consider Panel A which summarizes the evaluation of the strategies for a one year investment horizon. Notice that, for all strategies but the CPPI with $m = 6$, the minimum terminal value $\min V_T$ is higher than the guarantee ($G = 90$). For $m = 6$, the CPPI failed to ensure the guarantee exactly once, at Black Monday, October 19, 1987, with an excess return of -0.2046, i.e. a multiplier $m > |1 / -0.2046| = 4.8876$ induces a negative cushion. Although the average value of the time-varying multiple $m_{t,A2}$ is 4.49, and $m_{t,A2}$ varies between 0.0668 and 29.45, the worst case terminal portfolio value is ($\min V_T = 93.79$). Among the constant multiple strategies, the strategy with multiple $m^{*,\text{const}}$ indeed gives the highest average cushion growth rate. But, the growth rate is improved by all strategies with a time-varying multiple based on historical volatility estimates. The multiple $m_{t,A1}$ yields a 120% higher average cushion growth rate, the one based on the multiple $m_{t,A2}$ a 126% higher average cushion growth rate. This contradicts the assumption of a risk premium which is proportional to the squared volatility. At the same time, we conjecture

¹⁶Formally, the borrowing constraints are represented by an asset exposure $E_t = \min\{mC_t, wV_t\}$. $w - 1$ denotes the maximal fraction of the portfolio which can be financed by borrowing. If not mentioned otherwise, we use $w = 2$.

that the time-varying multiple strategies are able to outperform the standard CPPI. Intuitively it is clear that the growth rates of leveraged strategies are highly volatile. This is also the case here. For example, the standard deviation of the cushion growth rates is 0.36 (0.70) for the strategy $m_{t,A1}$ ($m_{t,A2}$) which is proportional to the inverse of the historical volatility (variance) estimate. Unsurprisingly, it turns out that none of the comparative growth rate results is significant.

We now comment on the ranking which is implied by the other performance measures. The Sharpe ratio (SR) is a mean-variance-based performance measure. If the investor values skewness positively, a mean-variance based measure overrates the strategies which reduce skewness (value strategies) and underrates strategies, which buy skewness (momentum or portfolio insurance strategies). *Ceteris paribus*, the Sharpe ratio is the lower the higher the leverage is. This is also observed in Panel A where the SR is the lower the higher the multiplier is. However, a ranking of the strategies along the lines of the Sharpe ratio is not meaningful in the case of PI strategies. Relying on the adjusted for skewness Sharpe ratio ($ASSR$) confirms a better performance of the time-varying multiple strategies compared to the optimal constant multiple strategy. This is also true for the omega measure.¹⁷ In addition, the time-varying multiple strategy are also ranked higher than the CPPI strategies according to the Sortino ratio (SoR) measure.¹⁸ Compared to the Sortino ratio, the so called upside potential ratio (UPR) uses as nominator the expected payoff of a call (with strike K) instead of the expected excess. However, the SoR also confirms the superiority of the strategies which rely on a time-varying volatility estimate. In summary, all performance measures indicate a superiority of the time-varying multipliers, excluding the Sharpe ratio.

Finally, it is worth to comment on the impact of the investment horizon on the performance measures. While Panel A refers to the performance evaluation w.r.t a yearly investment horizon, Panel B states the corresponding results assuming a monthly investment horizon. While a ranking according to the cushion growth rate is the same for both investment horizons, this is not true for the other performance measures. For example, in case of the Sortino ratio, the ranking of the constant multiplier strategies differs for the yearly and monthly horizon.

In summary, the results of Table 2 tend to favor the time-varying multiple strategies. The strategies which are based on the estimation of the daily forward volatility are promising candidates for outperforming the constant multiplier (CPPI) strategies w.r.t. the expected (cushion) growth rates as well as the other performance measures under consideration. However, as mentioned above, the descriptive results based on yearly non-overlapping historical return blocks do not allow the deduction of any significant performance results.

¹⁷The Omega measure accounts of the whole payoff distribution. According to a level K which defines the minimum acceptable payoff, the Omega measure divides the payoff into two parts. Note that we report $\Omega - 1$ throughout. This variation is proposed by Kazemi et al. (2004), who show that $\Omega - 1$ is negative if the expected excess $E[V_T - K]$ is negative. This allows an interpretation similar to the Sharpe Ratio.

¹⁸The SoR states the expected excess return (with respect to some level K) in relation to the square root of the second lower partial moment (semi-standard deviation, respectively). A theoretical foundation of this performance measure is given in Pedersen and Satchell (2002). The authors show the relation of the performance measure and the maximum principle.

With the exception of the benchmark static portfolio insurance strategy ($m = 1$), the strategies under consideration are leveraged strategies such that the standard deviation of the terminal values is rather high. In consequence, a significant answer to the research question if the performance of the optimal constant multiplier strategy can be improved by time-varying multiple strategies affords a sufficiently high number of observations (daily return paths, respectively). A meaningful solution is given in terms of a simulation model which mimics the empirical return distributions as close as possible.

4. SIMULATION MODEL

4.1. Model selection and estimation. Recall that our argumentation is, to a large extent, based on model risk. The *true* data generating process deviates from the assumed model. The investor is aware of this and optimizes her strategy in view of a whole model class instead of one particular model. However, the true data generating process is also not (perfectly) consistent with the benchmark assumptions on the relation between excess return and volatility, i.e. assumptions (A0), (A1), and (A2). It is also possible that the answer to the question of which of the assumptions on the risk premium is the most realistic one, depends to the market conditions itself, i.e. it can change over time. In order to be able to account of a rather complex interplay between expected excess return and volatility, we use a Markov Regime Switching Exponential GARCH-in-Mean (MRS-EGARCH-M) model. In addition, regime switching provides a feasible way to capture the observations of other empirical works which can be summarized as follows (cf. Whitelaw (2000)): Against the intuition the relation between conditional expected returns and conditional volatility is (i) often found to be weak or even negative, but (ii) this relation fluctuates significantly over time.¹⁹

In the following, we use a two-state Markov process and denote the regime indicator by $s_t = i$, $i = 1, 2$. The information set I_{t-1} contains information about all observed variables, particularly the excess returns $(R_{t-1}, R_{t-2}, \dots)$, but it does not contain the unobservable regime path $(s_{t-1}, s_{t-2}, \dots)$. The transition probabilities $\Pr(s_t = j | s_{t-1} = i) = p_{ji}$ are assumed to be constant and specified by the matrix

$$\begin{bmatrix} p_{11} & p_{21} \\ p_{12} & p_{22} \end{bmatrix} = \begin{bmatrix} p & (1-q) \\ (1-p) & q \end{bmatrix}. \quad (11)$$

The conditional distribution of excess returns is modeled as a mixture of two normal distributions, i.e.

$$R_t | I_{t-1} \sim \begin{cases} N(\lambda_{1,t}, \sigma_{1,t}^2) & \text{with probability } p_{1,t} \\ N(\lambda_{2,t}, \sigma_{2,t}^2) & \text{with probability } (1 - p_{1,t}), \end{cases} \quad (12)$$

where $\lambda_{i,t}$ denotes the state-dependent conditional expected excess return (the conditional risk premium) and $\sigma_{i,t}^2$ the conditional variance. In the literature, the probability $p_{1,t} = \Pr(s_t = 1 | I_{t-1})$ of being in regime 1 at time t is called *ex-ante* probability. The conditional variance $\sigma_{i,t}^2$ is given by an EGARCH model, cf. Nelson (1991). One advantage is the possibility of asymmetric responses of the volatility to positive and negative

¹⁹For a detailed information about regime-switching models with (E)GARCH volatility, we refer to the works of Gray (1996), Henry (2009), Klaassen (2002), Marcucci (2005), and Reher and Wilfing (2011).

shocks in the returns. In particular, the so-called *leverage effect* might differ over regimes. Furthermore, the exponential specification of σ_t^2 ensures a strictly positive volatility, i.e. without additional introduction of non-negativity constraints.

An important aspect concerns the dependence of the conditional excess returns on the volatility in each regime. For instance, we want to capture the possibility that $\lambda_{i,t}$ is constant in one regime but proportional to $\sigma_{i,t}$ or $\sigma_{i,t}^2$ in the other one. Comparing the goodness of fit of several specifications – also comprising variants proportional to $\ln \sigma_t^2$, i.e. variants which deviate from our three basic assumptions – it turns out that a combination of a constant c_i and a term proportional to volatility for both regimes, $\lambda_{i,t} = c_i + \theta_i \sigma_{i,t}$, gives the best description of the underlying data set.²⁰ For regime 1, the constant c_1 is negative and the volatility coefficient θ_1 is positive. In contrast, the opposite turns out for regime 2. In summary, we estimate the following model for the excess returns R_t by means of the ML-method:

$$\begin{aligned} R_t &= \lambda_{i,t} + \sigma_{i,t} z_t, & z_t &\sim N(0, 1), \\ \lambda_{i,t} &= c_i + \theta_i \sigma_{i,t}, \\ \ln \sigma_{i,t}^2 &= \omega_i + \alpha_i \left(|\tilde{z}_{i,t-1}| - \sqrt{2/\pi} \right) + \gamma_i \tilde{z}_{i,t-1} + \beta_i \ln \tilde{\sigma}_{i,t-1}^2. \end{aligned} \quad (13)$$

We specify the lagged variance $\tilde{\sigma}_{i,t-1}^2$ and the lagged shock $\tilde{z}_{i,t-1}$ entering the variance equation for state $s_t = i$ following Klaassen's approach. Klaassen (2002) suggests to integrate out the unobservable state at time $t-1$ by taking expectation of $\sigma_{i,t-1}^2$, conditional on the information that the current state is $s_t = i$. The lagged quantities and Klaassen's conditional probabilities $\tilde{p}_{ji,t-1}$ are computed as follows

$$\begin{aligned} \tilde{\sigma}_{i,t-1}^2 &= \tilde{p}_{ii,t-1} \sigma_{i,t-1}^2 + \tilde{p}_{ji,t-1} \sigma_{j,t-1}^2 + \tilde{p}_{ii,t-1} \tilde{p}_{ji,t-1} [\lambda_{i,t-1} - \lambda_{j,t-1}]^2, \\ \tilde{z}_{i,t-1} &= \tilde{p}_{ii,t-1} \frac{R_{t-1} - \lambda_{i,t-1}}{\sigma_{i,t-1}} + \tilde{p}_{ji,t-1} \frac{R_{t-1} - \lambda_{j,t-1}}{\sigma_{j,t-1}}, \\ \tilde{p}_{ji,t-1} &= \Pr(s_{t-1} = j | s_t = i, I_{t-1}) = \frac{p_{ji} \Pr(s_{t-1} = j | I_{t-1})}{\Pr(s_t = i | I_{t-1})} = \frac{p_{ji} p_{j,t-1}}{p_{i,t}}, \end{aligned}$$

with $i, j = 1, 2$. The ex-ante probabilities are calculated using the transition probabilities p_{ij} as defined by Equation (11) and the conditional probability density function $f(r_t | I_{t-1})$ corresponding to the conditional mixed-normal distribution of returns, cf. Equation (12), as

$$p_{j,t} = \Pr(s_t = j | I_{t-1}) = \sum_{i=1}^2 p_{ij} \left[\frac{f(R_{t-1} | s_{t-1} = i) p_{i,t-1}}{\sum_{k=1}^2 f(R_{t-1} | s_{t-1} = k) p_{k,t-1}} \right]. \quad (14)$$

Table 4 summarizes the quasi-maximum likelihood estimates of the model parameters. The asymptotic standard errors imply that all parameters are significantly different from zero. The smallest t-statistic (in absolute terms) is observed for c_1 with a value of 2.850. In order to guarantee stationarity of the conditional variance, the persistence parameter must

²⁰This is consistent with the literature. For example, Engle et al. (1987) suggest this form of the expected risk premium in a single regime (G)ARCH-in-Mean model. More recently, Reher and Wilfling (2011) observe similar results in a two-regime and more general GARCH-type model (nesting EGARCH as a special case).

Parameter estimates for the MRS-EGARCH model

	Regime 1	Asymp. SE		Regime 2	Asymp. SE
c_1	-0.001164	0.000409	c_2	0.002830	0.000278
θ_1	0.146237	0.049559	θ_2	-0.289444	0.041209
ω_1	-0.109877	0.012589	ω_2	-0.125000	0.036809
α_1	0.062288	0.006232	α_2	0.135578	0.016464
γ_1	-0.038539	0.005598	γ_2	-0.263822	0.012076
β_1	0.988012	0.001312	β_2	0.988012	0.003918
p	0.993919	0.001292	q	0.986645	0.002876

TABLE 4. Parameter estimates and asymptotic standard errors for the model along the lines Equation (13). The transition probabilities $p = p_{11}$ and $q = p_{22}$ are defined by Equation (11).

be restricted to $|\beta_i| < 1$ for both regimes. To obtain a stable simulation model with non-explosive volatility, we slightly sharpen this restriction and set an upper bound at $\beta_i \leq \beta$, where $\beta = 0.988012$ is the persistence parameter estimate obtained for a single-regime EGARCH model, *ceteris paribus*. We observe that both β_i are set to this upper bound by the estimation routine, i.e. volatility is highly persistent in both regimes.²¹ Also, the regimes are quite persistent as implied by the transition probabilities. Though, for regime 1 the estimated persistence is $1/(1 - p) = 164.45$ trading days. Regime 2 is estimated to persist just $1/(1 - q) = 74.88$ days. Accordingly, the unconditional probability of being in regime 1 (regime 2) is $(1 - q)/(2 - p - q) = 68.71\%$ (31.29%).

It is interesting to note that the simple notion of a high-mean, low volatility regime 1, and a low-mean, high volatility regime 2 is far from being true. In fact, for both regimes the conditional expected risk premium $\lambda_{i,t}$ can be positive or negative, depending on the level of conditional volatility. To be more precise, it holds $\lambda_{1,t} > 0$ if $\sigma_{1,t} > |c_1/\theta_1| = 0.00796$. In regime 2, the conditional mean $\lambda_{2,t}$ is positive if the volatility does not exceed the value 0.00978. The main difference in the variance equations across regimes concerns the response to shocks. Often, positive and negative shocks are interpreted in terms of good and bad news. In regime 1, the volatility $\sigma_{1,t}$ responses rather symmetrically to good and bad news, i.e. shocks of either sign increase the volatility.²² However, the impact of news is small. In sharp contrast, in regime 2 the volatility $\sigma_{2,t}$ is strongly increased after bad news, while good news slightly reduce volatility.

4.2. Additional benchmark strategies. The simulation model provides additional information according to the actual state $s_t = i$, $i = 1, 2$, and the actual volatility $\sigma_{i,t}$ and expected risk premium $\lambda_{i,t}$ that generate a return R_t at time t . Recall that the simulation model (data generating process) is not known to the investor. Nevertheless it is interesting to report the performance results of selected (visionary) strategies. Our first additional benchmark strategy $m_{t,\text{switch1}}$ is based on all information needed by a HARA investor to determine the myopic part of a strategy which is optimal in a diffusion setup (cf. Subsection 2.2) without the restriction to discrete time trading. In particular, the investor – which is from our point of view a visionary – knows the state s_t as well as the

²¹Further details on the model estimation, simulation and statistical tests are available from the authors on request.

²²This is because $|\alpha_1| > |\gamma_1|$, i.e. the magnitude effect dominates the leverage effect.

corresponding expected excess return $\lambda_{.,t}$ and the local volatility $\sigma_{.,t}$. For the special case that $\gamma = 1$, we denote the corresponding multiplier by $m_{t,\text{switch1}}$, i.e.

$$m_{t,\text{switch1}} = \begin{cases} \frac{\lambda_{1,t}}{\sigma_{1,t}^2} & \text{if } s_t = 1, \\ \frac{\lambda_{2,t}}{\sigma_{2,t}^2} & \text{if } s_t = 2. \end{cases} \quad (15)$$

The second (additional) benchmark strategy $m_{t,\text{switch2}}$ also uses the information about the current state s_t . However, the (partially visionary) investor does not know the local volatility $\sigma_{.,t}$ and the expected excess return $\lambda_{.,t}$. However, w.r.t. the set of multipliers comprising constant multipliers and multipliers proportional to the inverse of volatility or squared volatility, i.e. multipliers consistent with the benchmark assumptions (A0), (A1), and (A2), she knows which of these strategies performs best, given the regime. In particular, this strategy corresponds to the optimized (myopic) multiplier in a regime switching diffusion model setup as given in Subsection 2.2, cf. Equation (7). The corresponding strategy is denoted by $m_{t,\text{switch2}}$ where (for $\gamma = 1$) it turns out that

$$m_{t,\text{switch2}} = \begin{cases} 4 & \text{if } s_t = 1, \\ m_{t,A2} & \text{if } s_t = 2, \end{cases} \quad (16)$$

where $m_{t,A2}$ is defined as in Section 3, Equation (10), i.e. a multiplier proportional to the inverse of the simple rolling window estimate of conditional variance. Basically, this strategy (its performance results, respectively) sheds light on the question to what (maximal) extent a regime-dependent combination of the corner strategies corresponding to the basic assumptions on the risk premium can outperform the corner strategies themselves.

The third benchmark strategy is similar to $m_{t,\text{switch2}}$, but is not based on the knowledge of the regime/state. Instead, a Bayesian procedure is applied to infer the ex-ante probabilities $p_{i,t}$ (14) of the current state being 1 or 2. A similar Bayesian procedure is proposed in Hainaut and MacGilchrist (2012). The strategy $m_{t,\text{Bayes}}$ is defined in a similar manner as $m_{t,\text{switch2}}$, but the multiplier is averaged across $m = 4$ and $m_{t,A2}$ according to the regime probabilities, i.e.²³

$$m_{t,\text{Bayes}} = 4 p_{1,t} + m_{t,A2} p_{2,t}. \quad (17)$$

It is worth mentioning that this strategy is yet not admissible, because the inference of the regime probabilities is based on the knowledge about the model specification of $\lambda_{i,t}$ and $\sigma_{i,t}$. In addition, it also relies on the knowledge which corner strategies are the most promising, i.e. depending on the regime.

5. PERFORMANCE EVALUATION – SIMULATION RESULTS

The simulation tool established in the above section allows for significant results concerning the comparison of time-varying and constant multiple strategies. In addition, we account for the impacts of transaction costs. We investigate time-varying multipliers as a function of the volatility. The corner functions are fixed. Besides a constant multiple, the multiples $m_{t,A1}$ and $m_{t,A2}$ are proportional to the inverse of first and second power of the

²³Obviously, it is also possible to consider strategies that are not based on weighting the corner multipliers with the state probabilities. For example, an alternative and reasonable strategy is to base the decision on the most probable state. However, we restrict ourselves to the results of the strategy that performed best in our simulation setup.

Notation used for variable multiplier strategies

Corner strategy	Variable m_t proportional to the inverse of the:
$m_{t,A1}$	standard deviation of the latest 1 month (21 days) historical returns ($t, t-1, \dots, t-20$)
$m_{t,A2}$	variance of the latest 1 month (21 days) historical returns
Switching strategy	Information includes:
$m_{t,switch1}$	current regime, simulation model implied conditional expected excess return and volatility
$m_{t,switch2}$	current regime, best performing corner strategy (including constant mult.) given the regime
$m_{t,Bayes}$	best performing corner strategy (including constant mult.) given the regime, and ex-ante regime probabilities

TABLE 5. Summary of the dynamic multiplier strategies. $m_{t,A1}$ and $m_{t,A2}$ are consistent with assumptions (A1) and (A2), respectively, cf. Equations (9) and (10). In addition, the switching strategies include additional information: The strategy $m_{t,switch1}$ is based on the regime and the regime dependent myopic part. $m_{t,switch2}$ and $m_{t,Bayes}$ are, along the lines of Equation (7), proportional to a regime dependent power of the estimated current volatility. In the case of $m_{t,switch2}$, the regime is known. In contrast, $m_{t,Bayes}$ is based on a Bayesian inference of the ex-ante regime probabilities. For the initialization, we additionally use a 3 month set of pre-sampled returns, i.e. each year.

historically estimated standard deviation, i.e. proportional to the inverse of the standard deviation or variance, respectively, cf. Equations (9) and (10).²⁴ According to the intuition that for some cases it is better to take a constant multiple and for other cases (depending on market conditions) to introduce a multiple which is a non increasing function of the volatility, we also include switching strategies implied by the optimal multiplier of a regime switching diffusion setup, cf. Equation (7). In particular, the strategies are then proportional to the inverse of a power of the standard deviation where the power α_t itself depends on market conditions.

Recall that we do not assume that the true regime is known by the investor. The investor must thus decide how to estimate the probabilities of being in a certain regime. We consider a Bayesian approach giving rise to the strategy $m_{t,Bayes}$ where a detailed description is given in Subsection 4.1. Finally, we also include a visionary who knows the fraction of expected excess return and squared volatility in each regime. Along the lines of the degree of information, we denote the corresponding strategies by $m_{t,switch1}$ (regime and myopic multiplier is known) and $m_{t,switch2}$ (regime is known, but not the myopic multiplier). The basic features of the strategies are summarized in Table 5.

For expositional clarity, the simulation results are stated in two parts. In the first part, we abstract from transaction costs. We focus on the distribution of the variable multipliers, their daily changes and introduce some convenient key numbers which characterize the turnovers of the strategies. In the second part, we account for positive proportional transaction costs, introduce trigger trading (stochastic trading dates), and reconsider the evaluation of the PPI strategies for positive proportional transaction costs.

²⁴Probably, the results can be improved by means of more advanced volatility estimates. For example, a conditional variance model could be estimated on each path, i.e. on a simulated historic pre-sample. This issue is beyond the scope of the paper. It is left for further research. However, we conjecture that the gain of these strategies is rather small. The results which are generated by the strategies based on the simple estimates turn out to be quite close to those of the ones implied by the variance along the lines of the simulation model.

Summary statistics of variable multipliers m_t and the relative changes Δm_t

	Mean	Median	Stdev	Skewness	Kurtosis	Min
$m_{t,A1}$	2.857	2.228	2.307	4.617	36.809	0.066
$m_{t,A2}$	5.761	2.964	7.468	2.548	8.479	0.003
$m_{t,switch1}$	6.492	2.804	10.465	1.943	4.874	0.001
$m_{t,switch2}$	5.389	4.000	5.843	3.773	16.382	0.003
$m_{t,Bayes}$	5.504	3.907	5.548	3.797	16.696	0.010
$\Delta m_{t,A1}$	0.035	0.019	0.048	4.173	42.026	0
$\Delta m_{t,A2}$	0.067	0.035	0.099	6.444	137.653	0
$\Delta m_{t,switch1}$	0.218	0.008	1.787	27.459	894.201	0
$\Delta m_{t,switch2}$	0.032	0.000	0.231	34.719	967.030	0
$\Delta m_{t,Bayes}$	0.037	0.012	0.080	10.977	343.541	0

TABLE 6. Summary statistics of variable multipliers and absolute percentage daily changes $\Delta m_t = \left| \frac{m_t - m_{t-1}}{m_{t-1}} \right|$. The multipliers $m_{t,\cdot}$ are defined as in Table 5. According to a simple gap control, all multipliers are bounded by a maximum of $m_t = 32.3165$, which is equal to $|1 / -0.0309|$, where -0.0309 is the 1 % quantile of the dataset of daily excess returns. The results are approximated by a simulation of 50,000 years, each with 260 trading days.

5.1. CPPI and PPI evaluation in the absence of transaction costs. Table 6 summarizes the statistics of the dynamic multipliers and their relative changes. Recall that the optimal constant multiplier is $m^{*,\text{const}} = 1.7256$. On average, the variable multipliers are much larger. We also observe a high standard deviation (10.465) and a mean daily change of 21,8 % for the multiplier $m_{t,switch1}$, which might be explained by the fact that these multipliers are proportional to the model implied conditional risk premium $\lambda_{i,t}$, while all other strategies rely on a constant nominator $\hat{\lambda}$. In order to analyze the turnovers which are implied by the time-varying and constant multiplier strategies, we consider the following measures. Based on the relative daily turnover δ_t^S , given by

$$\delta_t^S := \frac{\left| m_t C_t - m_{t-1} C_{t-1} \frac{S_t}{S_{t-1}} \right|}{V_t}, \quad (18)$$

we report the expected maximum relative daily turnovers $Maxturn = E[\max_{t \in \{1, \dots, n-1\}} \delta_t^S]$, the expected total relative turnovers $Totturn = E[\sum_{t=1}^{n-1} \delta_t^S]$, and the expected number of trading days per year $Trades = E[\sum_{t=0}^{n-1} 1_{\delta_t^S > 0}]$.²⁵ If not mentioned otherwise, the results are based on a one year investment horizon ($T = 1$). As before, we assume $n = 260$ asset prices per year (corresponding to 259 simulated excess returns per path).

The performance results are summarized in Table 7. Basically, the descriptive results of Section 3 are confirmed. $m^{*,\text{const}}$ gives the highest cushion growth rate amongst the constant multiplier strategies. This is not true for all of the other performance measures. In contrast to the descriptive results, the simulation model gives a better description of the gap risk. We observe a rather high number of gap events for $m = 4$, $m = 6$, and $m = 8$. While CPPI strategies give rise to a high degree of gap risk, this is not the case for the time-varying strategies. There are no floor/guarantee violations observed for the multiplier

²⁵Notice that, for interpretability reasons, the key numbers $Maxturn$ and $Totturn$ do not account for the initial trading volume at $t = 0$. In contrast, this is the case for $Trades$. For example, the CPPI strategy with $m = 1$ and a positive initial cushion is a buy and hold strategy. It holds $Maxturn = 0$, $Totturn = 0$, and $Trades = 1$.

Performance results for dynamic and constant multiplier strategies

	$E[\ln \frac{C_T}{C_0}]$	$E[\ln \frac{V_T}{V_0}]$	$E[V_T]$	$\min V_T$	SR	$ASSR$	$\Omega - 1$	SoR	UPR	$Maxturn$	$Tottturn$
$m_{t,A1}$	0.219 <i>0.583</i>	0.047 <i>0.124</i>	105.751 <i>17.062</i>	91.202	0.337	0.493	7.793	3.739	4.219	0.156 <i>0.134</i>	3.228 <i>2.024</i>
$m_{t,A2}$	0.252 <i>0.787</i>	0.069 <i>0.167</i>	108.885 <i>23.355</i>	90.277	0.380	0.532	7.465	4.015	4.552	0.421 <i>0.251</i>	7.236 <i>3.584</i>
$m = 1$	0.033 <i>0.265</i>	0.006 <i>0.020</i>	100.602 <i>1.993</i>	90.000	0.302	0.279	1.299	0.477	0.844	0.000 <i>0.000</i>	0.000 <i>0.000</i>
$m^{*,const}$	0.036 <i>0.481</i>	0.011 <i>0.034</i>	101.124 <i>3.436</i>	90.000	0.327	0.338	1.480	0.609	1.021	0.005 <i>0.006</i>	0.249 <i>0.068</i>
$m = 4$	(-0.051) <i>1.133</i>	0.028 <i>0.090</i>	103.237 <i>10.331</i>	83.030 48	0.313	0.402	1.971	1.021	1.540	0.044 <i>0.024</i>	2.350 <i>0.733</i>
$m = 6$	(-0.225) <i>1.622</i>	0.041 <i>0.146</i>	105.495 <i>18.859</i>	75.721 206	0.291	0.380	2.251	1.339	1.934	0.109 <i>0.053</i>	5.556 <i>2.129</i>
$m = 8$	(-0.484) <i>2.058</i>	0.047 <i>0.181</i>	106.800 <i>24.172</i>	69.196 461	0.281	0.357	2.079	1.375	2.036	0.189 <i>0.089</i>	8.874 <i>3.861</i>
$m_{t,switch1}$	(0.559) <i>0.878</i>	0.127 <i>0.195</i>	115.976 <i>27.808</i>	89.721 2	0.574	0.832	20.288	8.006	8.400	1.403 <i>0.714</i>	14.464 <i>9.358</i>
$m_{t,switch2}$	(0.404) <i>0.766</i>	0.090 <i>0.170</i>	111.212 <i>24.153</i>	88.009 4	0.464	0.717	13.161	6.142	6.609	0.514 <i>0.422</i>	5.228 <i>2.802</i>
$m_{t,Bayes}$	(0.267) <i>0.821</i>	0.072 <i>0.165</i>	109.225 <i>23.030</i>	89.460 1	0.401	0.566	7.317	3.808	4.329	0.377 <i>0.296</i>	5.927 <i>3.440</i>

TABLE 7. Mean performance (*and standard deviations*) of the PPI strategies in the absence of transaction costs. The results are based on 50,000 simulations. The maximum investment quote is bounded to 200% of the current portfolio value V_t . The investment horizon is $T = 1$ year. The initial portfolio value is given by $V_0 = 100$. The guarantee is $G = 90$. The optimal constant multiplier is $m^{*,const} = 1.7256$. In line with the descriptive results in Table 2, the expected growth rates reported in parentheses are those where negative cushion pathes are excluded. The boldface number in the second line of the $\min V_T$ column counts the number of floor violations that occurred within the 50,000 simulations.

$m_{t,A1}$ and $m_{t,A2}$, i.e. the strategies which, independent from the current regime, reduce the multiple in times of volatile markets. In contrast, we observe some gap events for the switching strategies. Along the lines of reasoning given in Subsection 2.3, an additional gap control is needed for a final practical application of the strategies. However, in view of our main research question, it is worth mentioning that a gap control is more binding in the case of constant than time-varying multiples which reduce the multiplier in times of excess volatility. An additional gap risk control does not deteriorate the performance comparison presented here. An upper bound on the multiplier reduces the CE growth rates. The reduction is more severe if the gap control becomes binding more often. Next, it is important to observe that we validate the conjecture that the time-varying strategies proportional to inverse of the historical standard deviation and variance perform much better than the optimal constant multiplier. The out-performance turns out to be robust w.r.t. all performance measures. Obviously, the exact information w.r.t to the different regimes as used by the not feasible multiplier $m_{t,switch1}$ gives, by far, the highest average cushion growth (at least if 2 negative cushion pathes are not taken into account for). Intuitively, it is clear that the Bayesian switching multiplier $m_{t,Bayes}$ is also able to improve the performance. However, its average cushion growth rate is (only) 0.267 compared to 0.252 already obtained by $m_{t,A2}$. Unfortunately, the time-varying multiple strategies, in

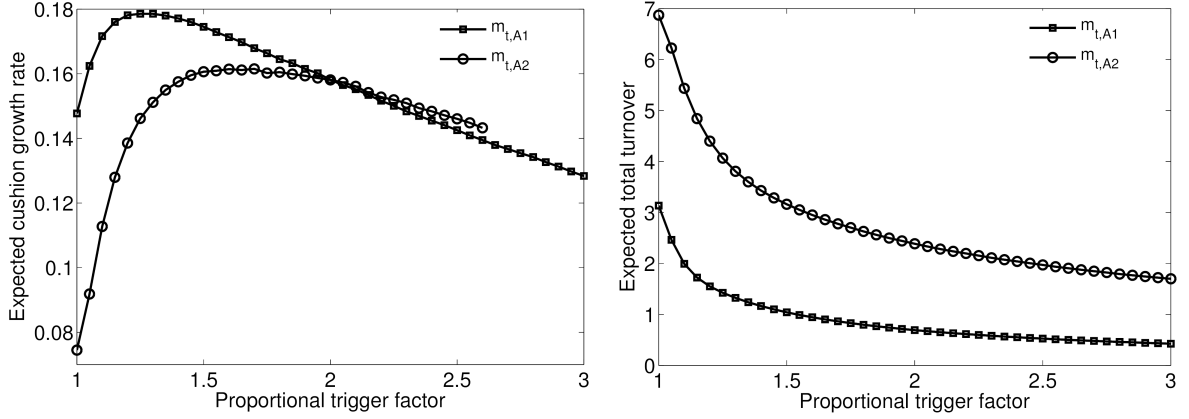


FIGURE 1. Left: Expected cushion growth rates for the dynamic multiplier strategies $m_{t,A1}$ and $m_{t,A2}$ and varying trigger levels specified by the proportional trigger factor φ and the trigger rule defined by Equation (19). The proportional transaction costs are given by $\theta = 0.25\%$. As the trigger factor φ increases from one (i.e. daily trading) to 3, the no-trade interval increases, too. For $m_{t,A2}$, we omit the results for $\varphi > 2.6$ where floor violations occurred. Right: Expected total relative turnovers $T_{\text{oturn}} = E[\sum_{t=1}^{n-1} \delta_t^S]$ where δ_t^S is defined by Equation (18) for the corresponding trigger levels. Results are based on 50,000 simulations.

particular the ones with the highest performance, afford high turnovers. Thus, it is necessary to reconsider the performance evaluation accounting for transaction costs and using adequate trigger designs.

Before we tackle the problem stemming from transaction costs, we reconsider the choice of $\gamma = 1$, i.e. the restrictions towards a log-utility in terms of the cushion. Recall that the guarantee component is rather high such that it is realistic to consider rather low values of γ . Table 8 depicts the certainty equivalent (CE) growth rates along the lines of Equation (3) for the benchmark strategies scaled according to $\gamma \in \{0.8, 1.0, 1.2, 1.4\}$. We consider two different investment horizons $T = 1$ year and $T = 10$ years, respectively. The results are given for capped multipliers (with borrowing constraints) and the uncapped counterpart strategies. First of all notice that the CE growth rates are, independent of the investment horizon, decreasing in γ , i.e. for all strategies. The higher the risk aversion is the lower is the CE growth rate. Intuitively, it is also clear that the loss due to borrowing constraints is the higher the lower the risk aversion is. However, our major concern is the utility loss and the distortion caused by neglecting the inter-temporal hedging demand linked to $\gamma \neq 1$. Obviously, a negative CE growth rate is prohibitive, in particular for an *optimized* multiplier. Notice that, for $\gamma = 1.4$, a negative CE growth rate is observed in the case of the *optimized* constant multiplier. The investor is even better off without any risky investment. In contrast, this is not the case for the time-varying multiples which still perform reasonably well. One might argue that neglecting the hedging demand even gives worse results in the case of a constant multiplier than a dynamic multiplier, i.e. a multiplier which is decreasing in the (local) volatility. However, the interpretation is to be taken with caution. In summary, we can nevertheless observe that the ranking of the strategies under consideration is not distorted when deviating from the assumption of $\gamma = 1$.

Certainty equivalent growth rates for different levels of risk aversion γ and investment horizons T

y^{CE}	<i>Capped investment quote</i>				<i>Uncapped investment quote</i>			
	$\gamma = 0.8$	$\gamma = 1.0$	$\gamma = 1.2$	$\gamma = 1.4$	$\gamma = 0.8$	$\gamma = 1$	$\gamma = 1.2$	$\gamma = 1.4$
Panel A: 1 year horizon								
$m_{t,A1}$	0.275	0.219	0.178	0.148	0.477	0.283	0.202	0.158
$m_{t,A2}$	0.301	0.252	0.212	0.179	1.398	0.645	0.386	0.266
$m^{*,\text{const}}$	0.061	0.036	0.019	-0.006	0.061	0.036	0.019	-0.006
Panel B: 10 year horizon								
$m_{t,A1}$	0.206	0.180	0.155	0.134	0.533	0.282	0.196	0.151
$m_{t,A2}$	0.225	0.203	0.181	0.160	1.654	0.642	0.366	0.250
$m^{*,\text{const}}$	0.064	0.037	0.017	-0.002	0.064	0.037	0.017	-0.002

TABLE 8. Certainty equivalent (CE) growth rates of the benchmark PPI strategies in the absence of transaction costs. The results are based on 50,000 simulations. The CE growth rates on the left hand side of the table refer to the case where the maximum investment quote is bounded to 200% of the current portfolio value V_t . On the right hand side, the corresponding CE growth rates of the otherwise identical uncapped strategies are quoted. The investment horizon is $T = 1$ year ($T = 10$, respectively). The initial portfolio value is given by $V_0 = 100$. The guarantee (subsistence level) is $G = 90$. The HARA parameter γ varies. The optimal constant multiplier is $m^{*,\text{const}} = \frac{1.7256}{\gamma}$.

5.2. Performance of triggered CPPI and PPI strategies under transaction costs. As indicated by the high turnovers of the strategies, accounting for transaction costs is especially important in the context of PPI strategies. The strategies imply a reduction of the asset exposure in falling markets. Analogously, the asset exposure is increased in rising markets. The investor suffers from any round-turn in the asset prices. The effect is particularly severe if there are in addition transaction costs, i.e. the effect is even leveraged by the transaction costs. Throughout the following, we consider proportional transaction costs denoted by θ . For daily trading, the cushion dynamics are given by Equation (8). In practice, the high turnovers of a strategy are normally controlled by implementing a trading filter. The trigger design is described as follows: Let τ denote a sequence of stopping times where $\tau_k \in \{0, 1, \dots, n-1\}$, $\tau_0 = 0$ and $\tau_{k+1} > \tau_k$. Assume that $C_{\tau_k+} > 0$ where C_{τ_k+} denotes that the transaction cost are already deducted, cf. Equation (8). The number of risky assets (constantly held immediately after τ_k) is $\eta_{\tau_k+} = \frac{m_{\tau_k} C_{\tau_k+}}{S_{\tau_k}}$ such that at t ($\tau_k < t < \tau_{k+1}$), the implicit multiplier is

$$m_t^{\text{imp}} = \frac{\eta_{\tau_k+} S_t}{C_t}.$$

In contrast, the target multiplier m_t is, at each trading date t , defined by the PPI rule. Along the lines of Paulot and Lacroze (2009), Hamidi et al. (2009a), MKaouar and Prigent (2007), Jessen (2009) and Khuman and Constantinou (2009), we only trigger the next rebalancing date if the deviations of the implicit multiplier and the target multiplier are

sufficiently high. We define

$$\tau_k := \inf \left\{ t > \tau_{k-1} \mid \left\{ m_t^{imp} \leq \frac{1}{\varphi} m_t \right\} \cup \left\{ m_t^{imp} \geq \varphi m_t \right\} \right\}, \quad (19)$$

where φ ($\varphi \geq 1$) denotes the proportional trigger factor. The special case that $\varphi = 1$ coincides with daily portfolio rebalancing.

Figure 1 illustrates the effects of varying trigger levels φ on the expected cushion growth rates. For illustrative purpose, we also depict the turnovers. Obviously, the expected total relative turnovers are the lower the higher the trigger level is. In contrast, the CE growth rate (expected cushion growth rate, respectively) is increasing in the trigger level for small trigger values but decreasing for high trigger values. The trade-off between more frequent rebalancing, i.e. sticking to the *optimized* PPI rule, and the transaction costs of the turnovers implies an optimal trigger level. Intuitively, it is clear that the optimal trigger depends on the PPI rule, i.e. the rule defining the multiple (cf. Balder et al. (2012)). Since our aim is the comparison of optimized PPI strategies, we use, for each strategy m , the (cushion growth rate) optimal level $\varphi^*(m)$.²⁶

Table 9 restates the results of Table 7 under transaction costs of $\theta = 0.25\%$.²⁷ Each strategy is evaluated w.r.t. its optimal trigger level φ^* . Observe that, for the constant multiplier strategies, the optimal level $\varphi^*(m)$ is the higher, the higher the multiplier is. The optimized trigger levels of the time varying strategies range from $\varphi^* = 1.25$ for $m_{t,A1}$ to $\varphi^* = 4.75$ for $m_{t,switch1}$. Intuitively, it is clear that the optimal trigger level is the higher, the higher the dispersion of the multiplier values is. Compared to the multiplier $m_{t,A1}$ proportional to the inverse of the volatility, a higher trigger level is observed for the multiplier $m_{t,A2}$ proportional to the inverse of the estimated variance. Using the expected cumulated relative turnovers *Totturn* caused by daily trading, cf. Table 7, as a benchmark, the optimized trigger trading gives a reduction. This is especially true for the time-varying multiplier strategies. Here, the reduction is more than 50%. It is clear that the trigger induces: (i) fewer trades because small (and probably negligible) deviations of the target multiple m_t and the implied multiple m_t^{imp} are not taken into account, and (ii) less turnovers in total. However, the average and maximum turnover volume per trade *Maxturn* increases. In practice, maximal turnovers which are above 50% are often considered as prohibitive. The highest values of *Maxturn* are ca 131% for the (model based) switching strategy $m_{t,switch1}$ and ca 62% for the multiplier $m_{t,A2}$. This is a major drawback of a time-varying multiple $m_{t,A2}$ proportional to the historically estimated variance, in particular if compared to the time-varying multiple $m_{t,A1}$ proportional to the historically estimated standard deviation.

²⁶Thus, the optimal trigger levels are estimated in sample. The procedure is merely justified by the aim to avoid a distortion of the ranking by an inadequate trigger level, i.e. a trigger design which is in favor of some strategies.

²⁷Proportional transaction costs of $\theta = 0.25\%$ constitute a rather high estimate – even for private investors – and may be interpreted as already containing transaction taxes present in many countries (and currently discussed in others). For instance, Jiang et al. (2009) assume $\theta = 0.18\%$, composed of 0.08% commission fees and 0.1% taxes for Chinese investors.

Performance results for dynamic and constant multiplier strategies under transaction costs and optimized trigger trading

	$E[\ln \frac{C_T}{C_0}]$	$E[\ln \frac{V_T}{V_0}]$	$E[V_T]$	$\min V_T$	$ASSR$	$\Omega - 1$	SoR	UPR	$Maxturn$	$Tottturn$	$Trades$
$m_{t,A1}$	0.179	0.041	105.034	91.099	0.449	5.922	2.984	3.487	0.204	1.426	19.769
$\varphi^* = 1.25$	<i>0.576</i>	<i>0.119</i>	<i>16.228</i>						<i>0.167</i>	<i>1.183</i>	<i>21.032</i>
$m_{t,A2}$	0.162	0.059	107.753	90.053	0.467	5.142	2.929	3.499	0.622	2.777	30.729
$\varphi^* = 1.70$	<i>0.832</i>	<i>0.164</i>	<i>22.731</i>						<i>0.348</i>	<i>1.702</i>	<i>43.186</i>
$m = 1$	0.031	0.006	100.576	90.000	0.269	1.225	0.454	0.825	0.000	0.000	1.000
$\varphi^* = 1$	<i>0.265</i>	<i>0.020</i>	<i>1.988</i>						<i>0.000</i>	<i>0.000</i>	<i>0.000</i>
$m^{*,const}$	0.034	0.010	101.091	90.000	0.330	1.436	0.590	1.000	0.010	0.038	5.440
$\varphi^* = 1.05$	<i>0.482</i>	<i>0.033</i>	<i>3.380</i>						<i>0.003</i>	<i>0.028</i>	<i>4.626</i>
$m = 4$	(-0.091)	0.023	102.756	82.537	0.350	1.581	0.844	1.378	0.052	1.211	48.690
$\varphi^* = 1.20$	<i>1.148</i>	<i>0.087</i>	<i>9.831</i>	45					<i>0.019</i>	<i>0.503</i>	<i>20.617</i>
$m = 6$	(-0.289)	0.034	104.546	73.547	0.328	1.745	1.069	1.682	0.135	2.428	42.260
$\varphi^* = 1.45$	<i>1.617</i>	<i>0.139</i>	<i>17.691</i>	204					<i>0.049</i>	<i>1.024</i>	<i>23.301</i>
$m = 8$	(-0.571)	0.037	105.549	60.559	0.301	1.586	1.076	1.754	0.246	3.682	48.882
$\varphi^* = 1.60$	<i>2.032</i>	<i>0.173</i>	<i>22.875</i>	461					<i>0.090</i>	<i>1.642</i>	<i>37.058</i>
$m_{t,switch1}$	(0.359)	0.089	111.260	87.255	0.651	11.098	5.030	5.483	1.309	7.850	49.701
$\varphi^* = 4.75$	<i>0.845</i>	<i>0.176</i>	<i>24.623</i>	1					<i>0.687</i>	<i>6.245</i>	<i>27.848</i>
$m_{t,switch2}$	(0.383)	0.082	110.174	81.748	0.682	13.423	6.046	6.496	0.549	2.489	50.504
$\varphi^* = 1.60$	<i>0.719</i>	<i>0.160</i>	<i>22.637</i>	4					<i>0.430</i>	<i>2.261</i>	<i>38.904</i>
$m_{t,Bayes}$	0.245	0.068	108.696	90.001	0.550	6.668	3.466	3.986	0.498	2.025	21.802
$\varphi^* = 1.55$	<i>0.819</i>	<i>0.161</i>	<i>22.336</i>						<i>0.350</i>	<i>1.902</i>	<i>34.302</i>

TABLE 9. Mean performance (*and standard deviations*) under proportional transaction costs of $\theta = 0,25\%$. The trigger design is described by Equation (19). The reported optimized proportional trigger factors φ^* are obtained numerically. To save space, the Sharpe ratio is not displayed. All other things (including the simulated set of paths) are identical to the setting of Table 7 in the above subsection. Note that *Trades* gives the expected number of trades per year, including the $t = 0$ trade for setting up the initial portfolio.

Concerning the performance evaluation under transaction costs and optimized trigger trading, we observe the following: Time-varying strategies which reduce the multiplier in times of volatile markets substantially outperform CPPI strategies. The result is robust for all performance measures under consideration. Finally, it is interesting to observe that the ranking of $m_{t,A1}$ and $m_{t,A2}$ is changed by the introduction of transaction costs. While $m_{t,A1}$ gives better results than $m_{t,A2}$ without transaction costs, the opposite is observed in the case of rather high transaction costs.

6. CONCLUSION

Recently, there is a growing trend to deviate from the concept of constant proportion portfolio insurance (CPPI) strategies. Instead of standardized constant multipliers, the industry relies on strategies with a rule based and time-varying multiple. We shed light on the question if the performance of constant proportion portfolio insurance (CPPI) strategies can be improved by means of such a time-varying multiplier.

Neglecting the inter-temporal hedging demand, the theoretical foundation of the strategies is given by maximizing the expected utility of a HARA investor in a diffusion model setup. Restricting the strategies to the myopic part of the multiplier implies the following

distinction: If the risk premium is assumed to be proportional to the squared volatility, the optimal strategy is a constant proportion portfolio insurance (CPPI) strategy. Otherwise, the multiple is time-varying.

We use S&P 500 index return data for the time period 1985-2013 to evaluate the performance of *optimized* portfolio insurance strategies. In addition to an optimal constant multiplier, we consider multipliers which are either proportional to the inverse of the estimated future volatility or the inverse of the squared volatility. Also, we account for strategies proportional to the inverse of a power of the volatility where the power itself depends on the market conditions (regimes, respectively). The main focus is on historical volatility estimates. Performance comparisons directly based on the empirical return series are presented and discussed. We circumvent the problems caused by a limited number of observations by introducing a simulation model. A regime switching EGARCH model also helps to shed light on the performance of dynamic PPI strategies which can distinguish between different regimes. Furthermore, the simulation model allows to account for the impacts of transaction costs and trigger trading on the performance evaluation.

It turns out that the current industry approach of a rule based and time-varying multiplier which is linked to a rolling window of historical volatilities significantly outperforms the (optimal) constant proportion portfolio insurance strategies. The out-performance is robust w.r.t. alternative performance measures and levels of risk aversion of the HARA investor. The result is still valid accounting for proportional transaction costs and gap risk, if an adequate trigger trading and gap control is implemented. However, it also turns out that the time-varying multiplier strategies can further be improved by accounting for regime switching strategies, i.e. strategies proportional to a power of the volatility where the power itself depends on the market conditions.

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